Changing velocity

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate acceleration and distinguish speeding up, slowing down, and turning.

SUCCESS CRITERIA - I can...

1. Calculate acceleration from velocity change and time.
2. Compare acceleration direction with velocity direction.
3. Explain why turning at steady speed is acceleration.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “acceleration” and “initial” to compare your model or result. Sentence starters are optional.

acceleration: rate of change of velocity

initial: at the beginning of the interval; subscript i

final: at the end of the interval; subscript f

01 Bring back an earlier idea

A velocity graph is constant at -4 m/s for 6 s.
Find displacement. A second object has components 8 m/s east and 3 m/s south; sketch its direction.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $a_{avg} = (v_f - v_i)/\Delta t$.
i = initial; f = final. Compare directions of velocity and acceleration to decide whether speed increases.

Worked reasoning: A cart moving east slows from 8 to 2 m/s in 3 s. Taking east positive, $a=(2-8)/3=-2$ m/s². It keeps moving east while acceleration points west.

Your turn: Draw initial and final velocity arrows and one acceleration arrow.
Use arrow lengths to show the speed change.

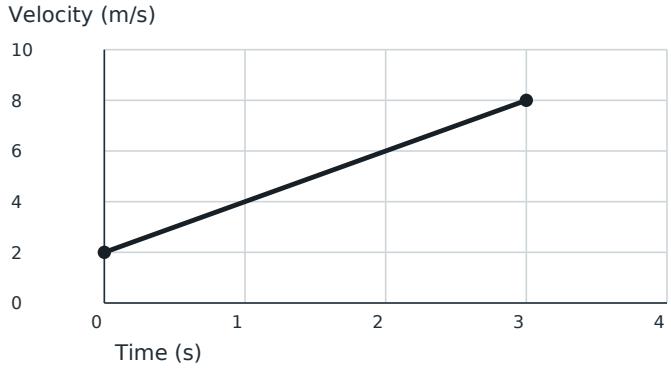
Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Watch velocity grow

Read the graph.
Find velocity change over 3 s, then acceleration. Is the cart speeding up?

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: ① ② ③

04 A bike slows

A bike moving east slows from 10 m/s to 4 m/s in 5 s. How much speed is lost each second? Which way does acceleration point?

Support: Step 1: Find the speed lost: 10 - 4. Step 2: Divide by 5 s. Step 3: Acceleration points opposite the bike’s motion while it slows.

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 A turn at steady speed

A toy car moves around a circle at steady speed. Does it accelerate?
Draw velocity arrows at two points to support your answer.

Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: 1 2 3

06 Choose a method

A goes 5 m east and 3 m north. B then goes 2 m west and 7 m south.
Draw the route and find the endpoint relative to the start.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

07 Exit check

A cart moves west at 6 m/s, then west at 2 m/s two seconds later. Is it speeding up or slowing down?
Find the acceleration direction and magnitude.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Make a velocity-time graph of a cart speeding up westward from 1 to 5 m/s in 2 s.
Use east as positive and explain the acceleration sign.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3